Polyhedron and Spectrahedron on fields with valuations

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Polyhedra and Spectrahedra over $\mathbb{R}$

For $\ell_1, \ldots, \ell_m \in \mathbb{R}[x]_{\leq 1}$ real affine n-variate functions:

Polyhedron: $\mathscr{P} = \{x \in \mathbb{R}^n : \ell_i(x) \geq 0, i = 1, \ldots, m\}$

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For $A_0, A_1, \ldots, A_n \in \mathbb{S}^m$ real symmetric matrices

Spectrahedron: $\mathscr{S} = \{x \in \mathbb{R}^n : A(x) = A_0 + x_1 A_1 + \cdots + x_n A_n \succeq 0\}$

"$\succeq 0$" = positive semi-definite = its eigenvalues are ≥ 0.
$A(x) \succeq 0$ is **called** a **LMI**.

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Fact: Polyhedra resp. Spectrahedra are **convex** as affine pre-images of resp.

$$\text{(nonnegative orthants)} \quad \mathbb{R}^m_+ = \{y \in \mathbb{R}^m : y_i \geq 0, i = 1, \ldots, m\}$$
$$\text{(positive semidefinite cones)} \quad \mathbb{S}^m_+ = \{X \in \mathbb{S}_m : X \succeq 0\}.$$

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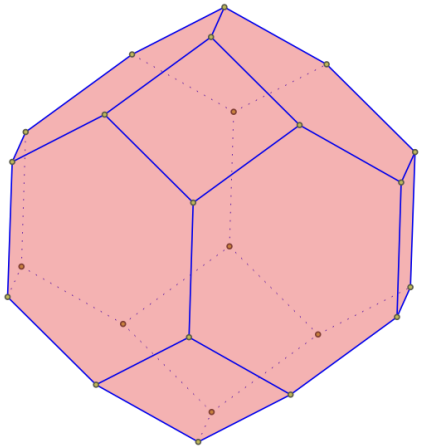
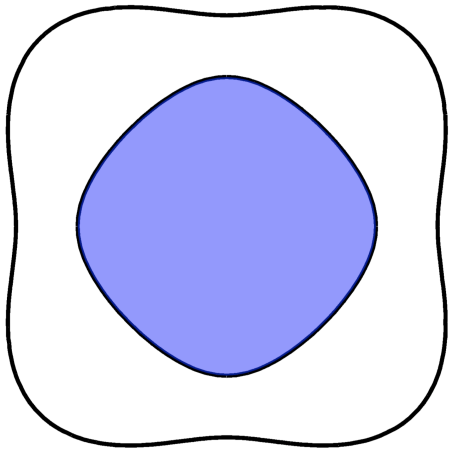
Fact: They are **basic semi-algebraic** sets:

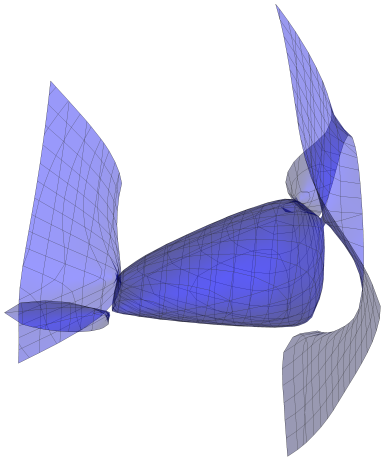
$$\det(A(x) + tI_m) = f_m(x) + f_{m-1}(x)t + \cdots + f_1(x)t^{m-1} + t^m$$

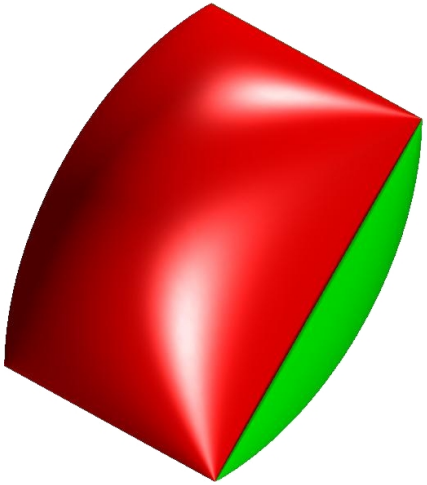
then $\mathscr{S} = \{x \in \mathbb{R}^n : f_i(x) \geq 0, i = 1, \ldots, m\}$.

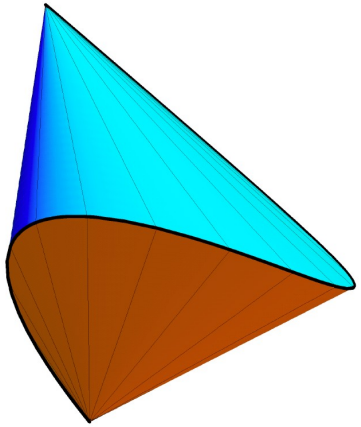
Polyhedra and Spectrahedra over $\mathbb{R}$



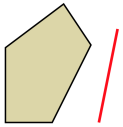


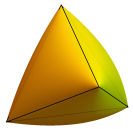






Motivations and properties

Polyhedra $\rightarrow$ feasible sets of Linear Programming (LP) 

Spectrahedra $\rightarrow$ feasible sets of Semidefinite Programming (SDP) 

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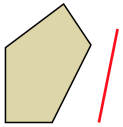
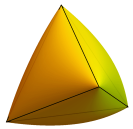
Polyhedra $\rightarrow$ feasible sets of Linear Programming (LP) 

Spectrahedra $\rightarrow$ feasible sets of Semidefinite Programming (SDP) 

Classical applications from

- real algebraic geometry (positive polynomials vs SOS, 17th Hilbert's problem)
- control theory (Lyapunov stability)
- combinatorial optimization (e.g. SDP relaxation of MAXCUT)
- polynomial optimization (moment-SOS relaxation hierarchy)

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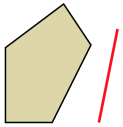
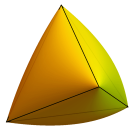
Polyhedra $\rightarrow$ feasible sets of Linear Programming (LP) 

Spectrahedra $\rightarrow$ feasible sets of Semidefinite Programming (SDP) 

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Several **good properties** besides convex

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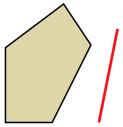
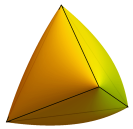
Polyhedra $\rightarrow$ feasible sets of Linear Programming (LP) 

Spectrahedra $\rightarrow$ feasible sets of Semidefinite Programming (SDP) 

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Several **good properties** besides convex but spectrahedra/SDP are highly more **pathological** than polyhedra/LP:

- spectrahedra are not closed under linear image
- solutions only over algebraic extension: $\{\sqrt{2}\} = \{x \in \mathbb{R} : \left[\begin{smallmatrix} x & 1 \\ 2 & x \end{smallmatrix}\right] \succeq 0 \text{ and } \left[\begin{smallmatrix} 1 & x \\ x & 2 \end{smallmatrix}\right] \succeq 0\}$
- in SDP: weak infeasibility, not attainement of inf/sup, etc...

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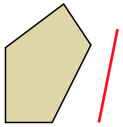
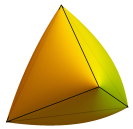
Polyhedra $\rightarrow$ feasible sets of Linear Programming (LP) 

Spectrahedra $\rightarrow$ feasible sets of Semidefinite Programming (SDP) 

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Semidefinite representable sets

The linear projection of a spectrahedron is called a spectrahedral shadow:

$$S = \left\{ x \in \mathbb{R}^n : \exists\, y \in \mathbb{R}^p, A_0 + \sum_{i=1}^{n} x_i A_i + \sum_{j=1}^{p} y_j B_j \succeq 0 \right\},$$

for matrices $A_i, B_j \in \mathbb{S}^d$.

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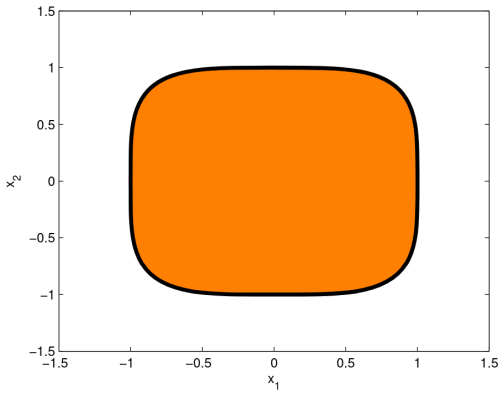

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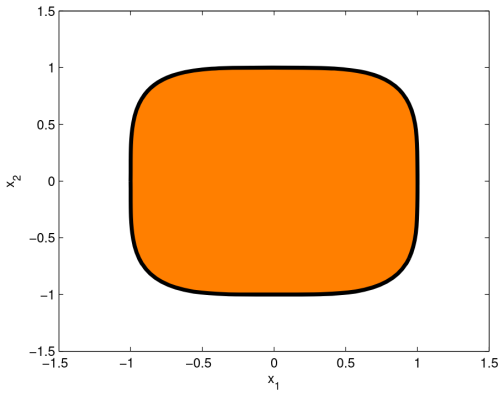

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Not every convex (compact) semialgebraic set is semidefinite representable (Scheiderer, 2018).

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Setting

Discrete-valuation fields

Let K be a field with:

- discrete valuation $v_K : K \to \mathbb{R} \cup \{+\infty\}$ making it complete,
- integer ring O_K,
- residue field κ,
- uniformizer π, whose valuation generates the additive group $\mathrm{val}(K)$.

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Example

1. $\mathbb{Q}_p$, p-adic valuation, $\mathbb{Z}_p$, $\kappa = \mathbb{F}_p$, $\pi = p$.
2. $\mathbb{F}_q((t))$, t-adic valuation, $\mathbb{F}_q[[t]]$, $\kappa = \mathbb{F}_q$, $\pi = t$.
3. $\mathbb{Q}((t))$, t-adic valuation, $\mathbb{Q}[[t]]$, $\kappa = \mathbb{Q}$, $\pi = t$.
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Topology somehow in-between discrete and continuous mathematics.

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Non-archimedean balls

Definition (Ball/disks)

$$D_K(\alpha;\lambda) := \{x \in K : \ v_K(x-\alpha) \geq \lambda\}.$$

Definition (Polydisks)

$$D_{K^n}(\alpha_1,\ldots,\alpha_n;\lambda_1,\ldots,\lambda_n) := \{x \in K^n : \ \forall i, \ v_K(x_i-\alpha_i) \geq \lambda_i\}.$$

Proposition

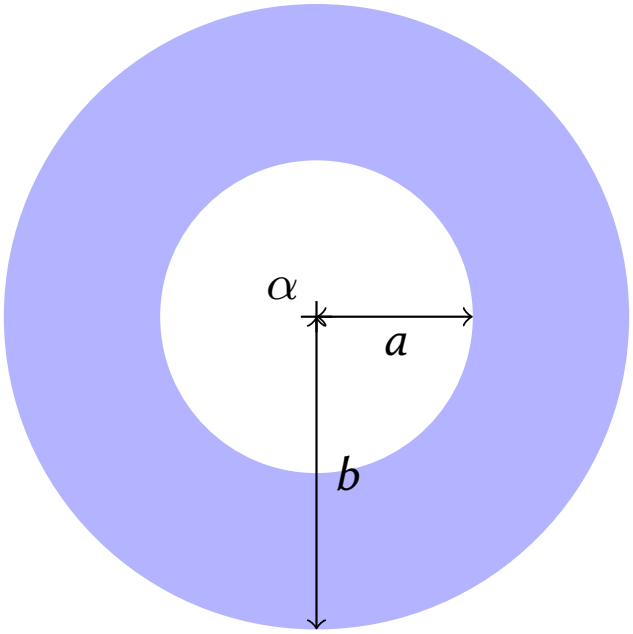
- *Any point of $D_K(\alpha;\lambda)$ is a center of this disk.*
- *If two disks have a non-trivial intersection, then one contains the other. As such, any set of disks as a forest / tree-like structure for inclusion.*

Annuli

Definition (Annulus)

$$\mathcal{C}_K(\alpha, a, b) = \{x \in K \mid a \leq \mathrm{val}(x - \alpha) \leq b\},$$

for $x \in K$, $a \leq b \in \mathbb{R}$.



Remark

Central object in p-adic differential equations or in non-archimedean analytic geometry.
Used to define series converging on $\mathcal{C}_K(\alpha, a, b)$.

Discoids

Definition

Let $\varphi \in K[x]$ be a monic polynomial and let $\lambda \in \mathbb{Q}$. Then

$$D_K(\varphi; \lambda) := \{x \in K : \ v_K(\varphi(x)) \geq \lambda\}$$

is the **discoid** with center φ and radius λ over K.

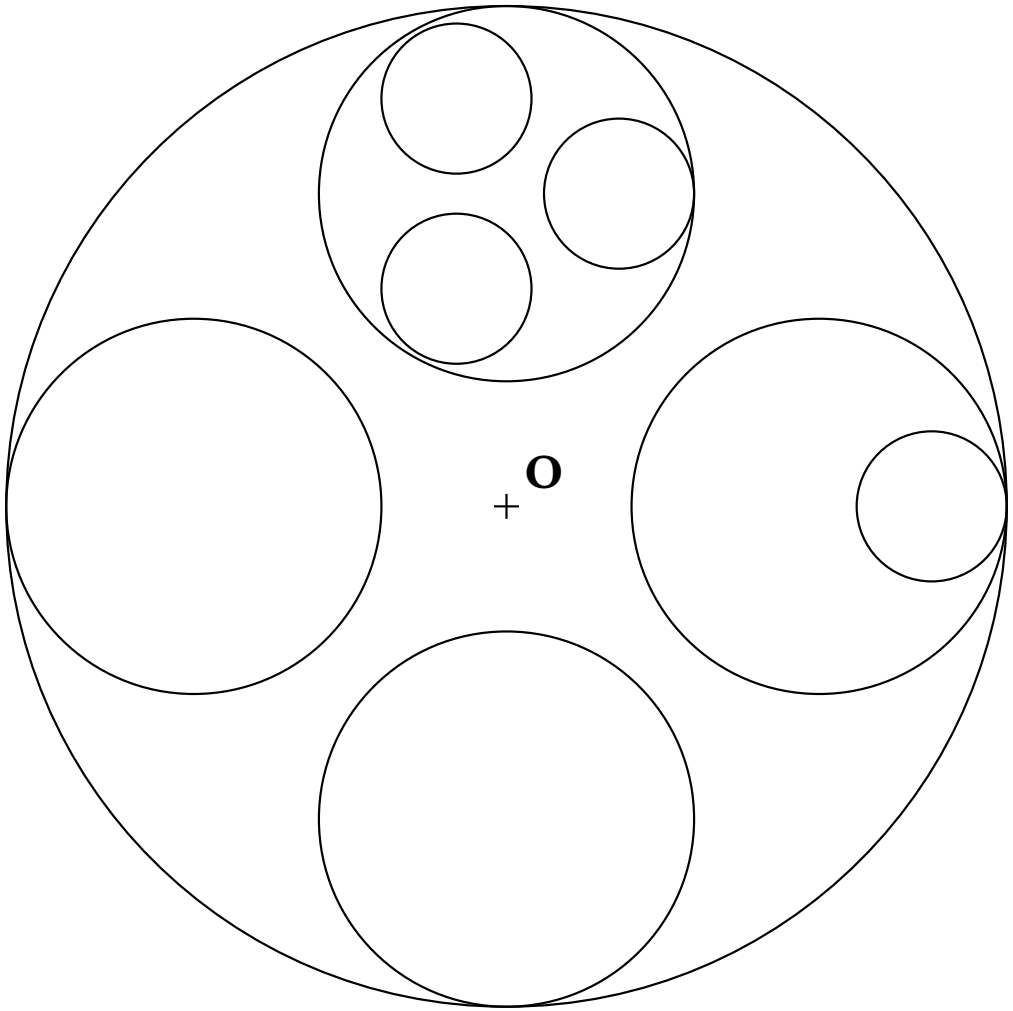
Remark

Discoids play a central role in the family of OM factorization algorithms.
Thanks to recent works with A. Poteaux and M. Weimann, they appear in the clusterization of the roots of polynomials over fields with valuations.

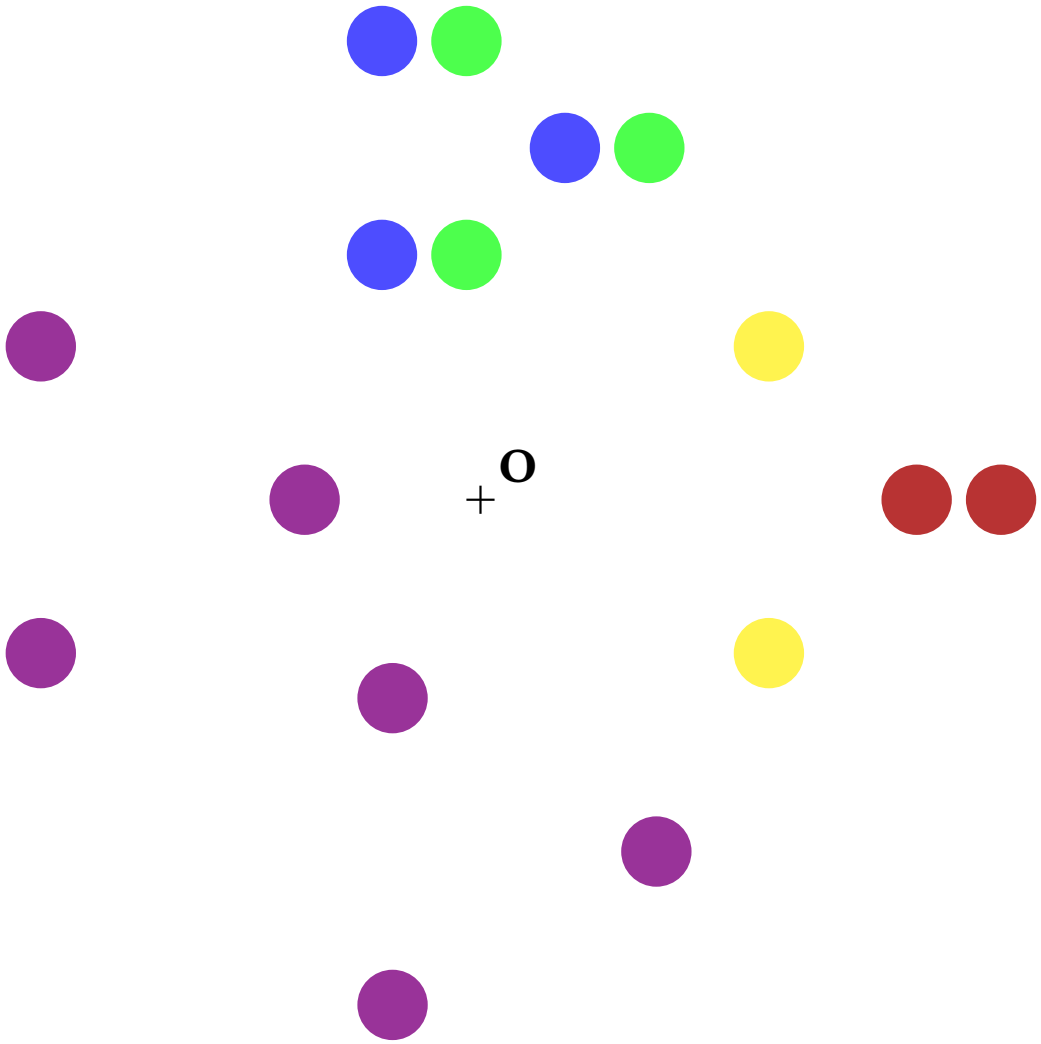
Visualizing discoids

Proposition

If φ is irreducible over K, then $D_{\overline{K}}(\varphi;\lambda)$ is the union of balls of same radius centered around the roots of φ. The Galois group of $\overline{K}/K$ acts on those balls.



Visualizing clustering roots in a discrete-valuation field with diskoids



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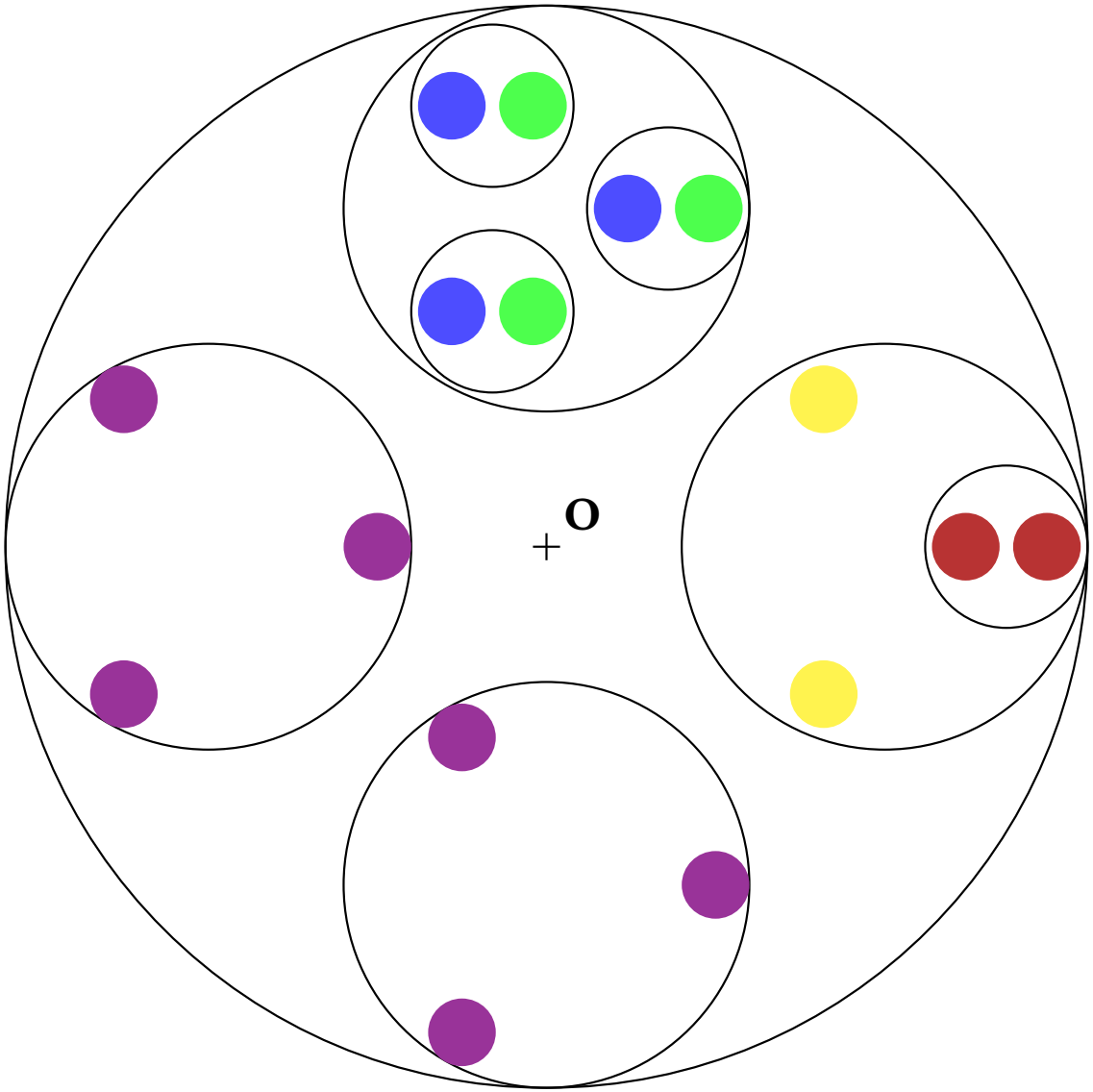
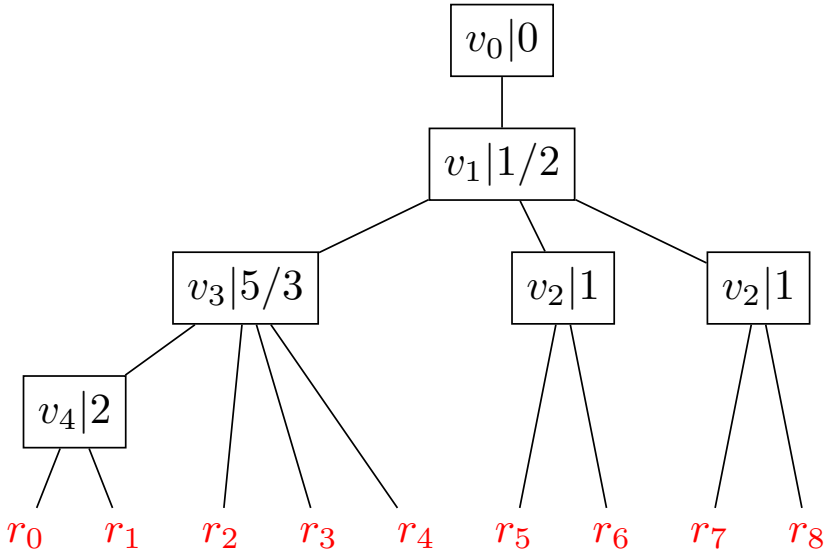


Clustering roots with discoids and OM algorithm: over $\mathbb{Q}((t))[x]$



Berkovich skeleton of $f_1 = (x - t^2)(x - 2t^2)(x^3 - t^5)((x^2 - t)^2 - t^3)$

In black: Balls of diskoids defined by:

v_0: $D(0, 0)$.

v_1: $D(\Phi_1, \frac{1}{2})$, with $\Phi_1 = x$.

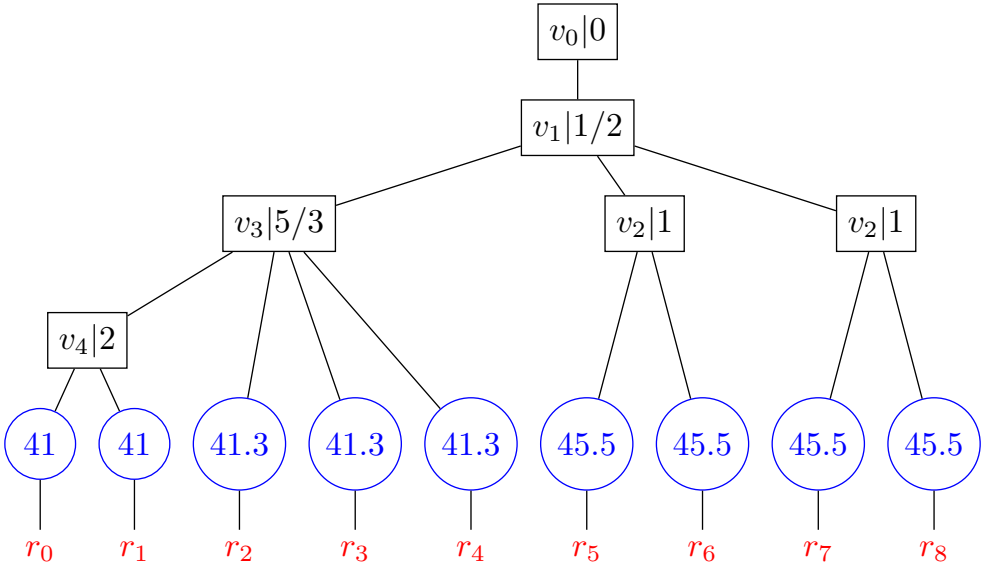
v_2: $D(\Phi_2, 3/2)$, with $\Phi_2 = \Phi_1^2 - t$.

v_3: $D(\Phi_3, \frac{5}{3})$, with $\Phi_3 = x$.

v_4: $D(\Phi_4, 2)$, with $\Phi_4 = x$.

In red: Roots.

Clustering roots of approximation of polynomials in $\mathbb{Q}((t))[x]$



Approximate Berkovich skeleton clustering all the roots of

$$f_1 = (x - t^2)(x - 2t^2)(x^3 - t^5)((x^2 - t)^2 - t^3) + O(t^{50})$$

In black: Balls of diskoids defined by:

v_0: $D(0, 0)$.

v_1: $D(\Phi_1, \frac{1}{2})$, with $\Phi_1 = x$.

v_2: $D(\Phi_2, 3/2)$, with $\Phi_2 = \Phi_1^2 - t$.

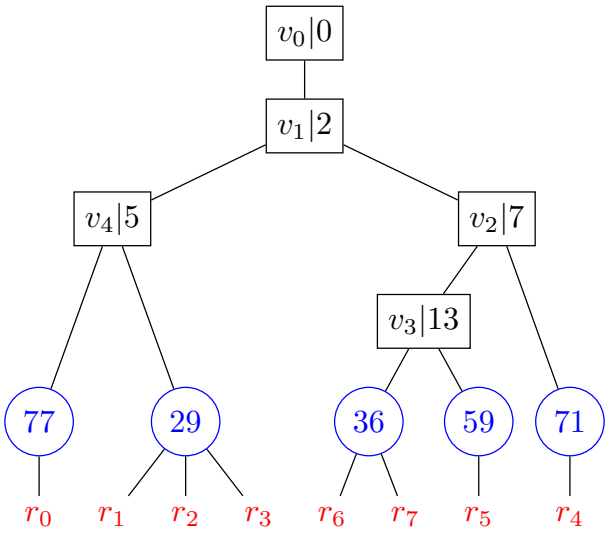
v_3: $D(\Phi_3, \frac{5}{3})$, with $\Phi_3 = x$.

v_4: $D(\Phi_4, 2)$, with $\Phi_4 = x$.

In blue: Balls coming from Brink's inequality.

In red: Roots.

Clustering roots of approximation of polynomials in $\mathbb{Q}((t))[x]$, multiplicity



Approximate Berkovich skeleton clustering all the roots of

$$f_3 = (x - t^2)(x - t^2 - t^5)^3(x + t^2 + t^7)^2(x + t^2 + t^7 + t^{13})(x + t^2) + O(t^{100})$$

In black: Balls of given log-radii defined by diskoids defined by the following valuations:

v_0: $D(0, 0)$

v_1: $D(0, 2)$

v_2: $D(t^2, 7)$

v_3: $D(t^2 + t^7, 13)$

v_4: $D(-t^2, 5)$

In blue: Balls coming from Brink's inequality.

In red: Roots.

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Non-archimedean polyhedron

Definition

A *polyhedron* is a subset $\mathcal{P} \subset K^n$ of the form

$$\mathcal{P} = \left\{ x \in K^n : \begin{array}{ll} \mathrm{val}(\ell_i(x)) \geq 0 & i = 1, \ldots, d \\ \mathrm{val}(m_j(x)) = +\infty & j = 1, \ldots, e \end{array} \right\},$$

for some $\ell_1, \ldots, \ell_d, m_1, \ldots, m_e \in K[x]_1$ (polynomials of degree 1).

This can be written in matrix form:

$$\mathcal{P} = \{x \in K^n : Ax + v \succeq 0, Bx + w = 0\},$$

for $A \in K^{d \times n}$, $v \in K^d$, $B \in K^{e \times n}$, and $w \in K^e$ such that $Ax + v = (\ell_1, \ldots, \ell_d)^T$ and $Bx + w = (m_1, \ldots, m_e)^T$.

Example

$$D(c, \alpha) = \{x \in K^n : val\,(x - c) \geq \alpha =\} = \left\{x \in K^n : val\left(\pi^{-\alpha}(x - c)\right)\right\}.$$

As such, balls are polyhedron, and similarly for polydisks.

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Projecting polyhedron

Theorem (direct image (DI))

Let $f : K^n \to K^m$ be an affine map and $\mathcal{P}$ a polyhedron in K^n. Then $f(\mathcal{P})$ is a polyhedron in K^m.

Idea of proof.

Use the Smith Normal Form of the linear part f_0 of f.

$$Mat_B(f_0) = P \begin{pmatrix} \pi^{\alpha_1} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \pi^{\alpha_2} & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \pi^{\alpha_s} & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix} Q$$

with $P, Q \in GL_n(O_K)$.
Reduce to the case of: translation, isometry, projection. $\square$

Applications

Proposition (Characterization of polyhedra)

A non-empty polyhedron is the image of a polydisc under an affine map.

Proposition

Polyhedra in K are either the empty set or balls, possibly with infinite radius.

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Definition and characterization

Definition

We say that a matrix $M \in K^{d\times d}$ is positive semidefinite ($M \succeq 0$) if

$$\mathsf{Spec}_{\overline{K}}(M) \subset O_{\overline{K}}.$$

We denote by $K_+^{d\times d} \subset K^{d\times d}$ the set of positive semidefinite matrices.

Theorem

$M \in K_+^{d\times d}$ if and only if $\chi_M \in O_K[T]$.

Proof.

Use Newton polygons. $\square$

Corollary

$$O_K^{d\times d} \subset K_+^{d\times d}.$$

(Counter)-Example

Example

Let

$$M = \begin{bmatrix} \frac{3}{5} + 5 & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix} \in \mathbb{Q}_5)^{2\times 2}.$$

Then

$$\chi_M(T) = T^2 - 5T - 4,$$

Hence

$$M \in (\mathbb{Q}_5)_+^{2\times 2}$$
$$M \notin (\mathbb{Z}_5)^{2\times 2}.$$

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What is a spectrahedron

Definition

A *spectrahedron in* $K^{d\times d}$ is a set of the form

$$\mathcal{S} = \mathcal{L} \cap K_+^{d\times d} = \left\{ X \in K^{d\times d} : \exists\, x \in K^n, X = A_0 + \sum_{i=1}^{n} x_i A_i, X \succeq 0 \right\}$$

where $\mathcal{L} = A_0 + \mathrm{span}(A_1, \ldots, A_n) \subset K^{d\times d}$ is an affine space, for some $A_0, A_1, \ldots, A_n \in K^{d\times d}$.

Definition

A *spectrahedron in* K^n is any affine section of a set of the form

$$\mathcal{S}_A = \varphi_A^{-1}(K_+^{d\times d}) = \{x \in K^n : A_0 + x_1 A_1 + \cdots + x_n A_n \succeq 0\}$$

for some $d \in \mathbb{N}$, where $\varphi_A : K^n \to \mathcal{L} \subset K^{d\times d}$ is the map $\varphi_A(x) = A_0 + x_1 A_1 + \cdots + x_n A_n$, and $\mathcal{L} = A_0 + \mathrm{span}(A_1, \ldots, A_n)$.

Polyhedron implies Spectrahedron

Proposition

A polyhedron in K^n is a spectrahedron in K^n.

Proof.

Assume $\mathcal{P} = L \cap M$ be a polyhedron in K^n, with $L = \{x \in K^n : \mathrm{val}\ell_i(x) \geq 0, i = 1, \ldots, d\}$ and $M = \{x \in K^n : \mathrm{val}m_j(x) = +\infty, j = 1, \ldots, e\}$, for some $d, e \in \mathbb{N}$. Let $A_0, A_1, \ldots, A_n \in K^{d \times d}$ be such that

$$\varphi_A(x) := A_0 + x_1 A_1 + \cdots + x_n A_n = \mathtt{diag}(\ell_1(x), \ldots, \ell_d(x)).$$

One has $L = \mathcal{S}_A = \varphi_A^{-1}(K_+^{d \times d})$. Since $M \subset K^n$ is affine, then $\mathcal{P}$ is a spectrahedron in K^n. $\square$

Some spectrahedron are not polyhedron

Example

The following spectrahedron is not a polyhedron:

$$\mathcal{S} = \left\{ x \in K : \begin{bmatrix} 0 & x + \pi^{-1} \\ x & 0 \end{bmatrix} \succeq 0 \right\}.$$

Proof.

By characterization of positive semi-definite matrices, one has
$\mathcal{S} = \{x \in K : \mathrm{val}(x(x + \pi^{-1})) \geq 0\} = \{x \in K : \mathrm{val}x \geq 1\} \cup \{x \in K : \mathrm{val}(x - \pi^{-1}) \geq 1\}$
which is not a ball in K. Therefore, $\mathcal{S}$ is not a polyhedron. $\square$

Spectrahedron in K

Proposition

Spectrahedra of K are diskoids.

Proof.

If S is a spectrahedra of K not reduced to a point, then

$$\begin{aligned} S &= \{x \in K : A_0 + xA_1 \succeq 0\} \\ &= \cap_{i=0}^{d-1} \{x \in K : \mathrm{val}(a_i(x)) \geq 0\} \end{aligned}$$

for some integer d, matrices A_0 and A_1 in $K^{d \times d}$, The intersection of two diskoids is either empty or a diskoid. Consequently, spectrahedra of K are diskoids. □

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Spectrahedral shadows

Definition (Semidefinite-representable set)

A set $\mathcal{R} \subset K^n$ is called *semidefinite-representable (SDR)* if there exist $m, d \in \mathbb{N}$ and a spectrahedron $\mathcal{S} \subset K^{n+m}$ such that

$$\mathcal{R} = \mathtt{proj}_{\leq n}(\mathcal{S})$$

with $\mathtt{proj}_{\leq n} : K^{n+m} \to K^n$ the projection on the first n components.

If $m \geq 1$, we can also write that $\mathcal{R}$ is a spectrahedral shadow.

Annuli and shadows

Theorem

Annuli in K such as $\mathcal{C}_K(0,a,b) = \{x \in K \mid a \leq \mathrm{val}(x) \leq b\}$ are spectrahedral shadows.

Proof.

$$M(x,y) := \left(\begin{bmatrix} \pi^{-a}x & 0 & 0 & 0 \\ 0 & \pi^{b}y & 0 & 0 \\ 0 & 0 & \pi^{-1} & -\pi^{-1}x \\ 0 & 0 & \pi^{-1}y & -\pi^{-1} \end{bmatrix} \right) \succeq 0.$$

Then $M \succeq 0$ if and only if (x,y) is a solution of the system of inequalities:

$$\begin{cases} \mathrm{val}(\pi^{-a}x) \geq 0 \\ \mathrm{val}(\pi^{b}y) \geq 0 \\ \mathrm{val}(\pi^{-1} - \pi^{-1}) = +\infty \geq 0 \\ \mathrm{val}(\pi^{-2}(xy-1)) \geq 0 \end{cases} \iff \begin{cases} \mathrm{val}\,(x) \geq a \\ \mathrm{val}\,(y) \geq -b \\ \mathrm{val}\,(xy-1) \geq 2 \end{cases}$$

Thus $\mathrm{val}\,(x) + \mathrm{val}\,(y) = \mathrm{val}\,(xy) = \mathrm{val}\,(-1) = 0$. Therefore $\mathrm{val}\,(x) = -\mathrm{val}\,(y) \leq -(-b) = b$ and $x \in \mathcal{C}_K(0,a,b)$. The converse is clear and $\mathcal{C}_K(0,a,b)$ is the proj. of the spectrahedra defined by M. $\square$

Annuli and spectrahedron

Theorem

If the residue field of K is infinite, no (non-trivial) annulus of K can be a spectrahedron of K.

Proof.

Spectrahedron of K are diskoids and thus are finite union of balls.
If the residue field of K is infinite, no annulus can be finite union of balls.
For example:

$$\{x \in K \mid \mathrm{val}(x) = 0\} = \amalg_{l \in \kappa} B(b_l, |\pi|)$$

for some representatives $b_l \in O_K$ of the classes of $O_K/\pi O_K = \kappa$. It thus can not be covered by a finite amount of balls of radius $|\pi|$ and if one take any ball of radius 1 centered at a point of the sphere, then one gets the whole $B(0, 1)$.
One can generalize this argument to annuli. $\square$

Conclusion

Summary

- Definition of polyhedron and spectrahedron over K.

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Future works

- Can we decide if a spectrahedron is non-empty?

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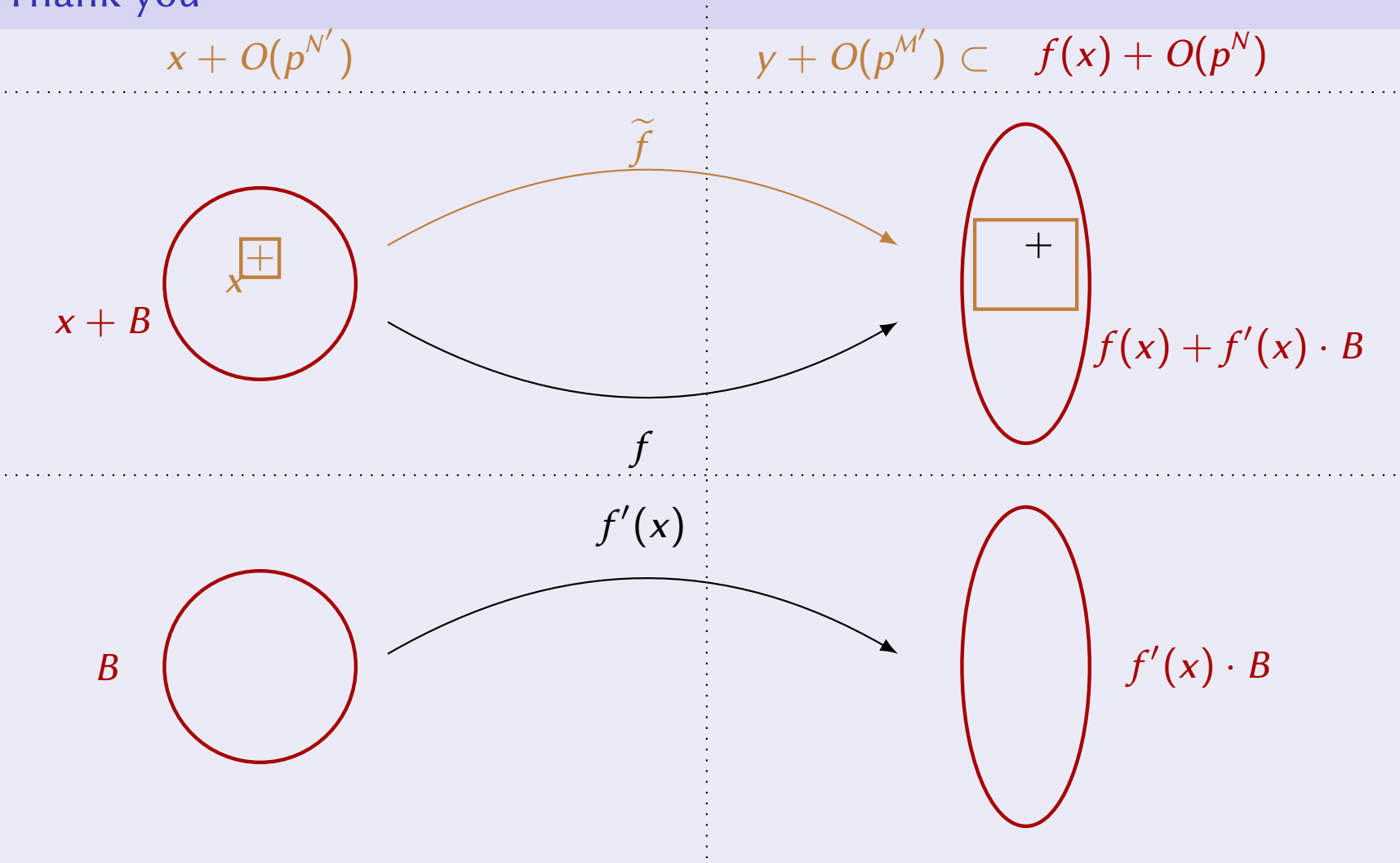
Thank you for your attention

Thank you



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